

Yansong Qu

Ph.D. Student in Transportation Engineering | Autonomous Driving & Intelligent Systems

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RESEARCH INTERESTS

Autonomous driving; vision-language-action models; multimodal large language models; world models; reinforcement learning; multi-agent systems; spatiotemporal learning; traffic safety.

EDUCATION

Purdue University, West Lafayette, IN

2024–Present

Ph.D. in Civil Engineering (Transportation) | GPA: 3.9/4.0 | Expected: Dec 2027

Advisor: Prof. Samuel Labi; **Co-advisor:** Prof. Sikai Chen. Research areas: LLMs for autonomous driving and LLM-enhanced reinforcement learning.

Selected coursework: Reinforcement Learning, Introduction to Robot Learning, Motion Planning, Imitation Learning and Robotics, and Vehicular Cyber-Physical Systems.

Beijing University of Technology, Beijing, China

2021–2024

M.S. in Transportation Engineering | GPA: 3.56/4.0 (87.78/100), Top 5%

Dalian Jiaotong University, Dalian, China

2017–2021

B.S. in Transportation Engineering | GPA: 3.47/4.0 (84.56/100), Top 10%

EXPERIENCE

Research Intern, Honda Research Institute (HRI)

May 2026–Aug 2026

- Built a pipeline for processing large-scale multimodal autonomous-parking data, standardizing raw records into datasets reusable across perception, planning, intent prediction and chain-of-causation (CoC) tasks.
- Developed tooling for multi-track label generation and human correction of perception labels and behavioral annotations, improving the consistency and repeatability of data quality control and verification.

Research Assistant & Teaching Assistant, Purdue University

Aug. 2024–Present

- Design and implement autonomous-driving models, conduct experiments, analyze results, and contribute to research manuscripts.
- Serve as teaching assistant for CE 39800, *Introduction to Civil Engineering Systems Design*; lead tutorials, mentor team design projects, and evaluate assignments.

SPONSORED RESEARCH PROJECTS

World Models for Deployable Reinforcement Learning Agents: Autonomous Driving Across

2026

Simulation and Physical Domains

- Pretrain representation-centric latent world models on multi-source driving data (CARLA, replay simulators, and real rally/AV logs) to learn controllable long-horizon dynamics and support policy learning through imagination.
- Train single- and multi-agent policies in learned latent environments using imagination-based RL, with decentralized control coordinated by a centralized critic for cooperative, safety-critical behavior.
- Develop a unified CARLA-to-ROS 2 sim-to-real pipeline for rally-car and full-scale AV testbeds, incorporating domain adaptation, safety monitors, and standardized benchmarks.

Towards Safe and Efficient Autonomous Driving: Human Expertise and Multimodal Large Language Models

2025

- Design a dual-stage framework that integrates multimodal LLMs and human expertise to improve autonomous-driving safety in rare and safety-critical scenarios.
- Fuse vision, language, and action modalities with structured reasoning for interpretable sequential decision-making and real-time low-level control.
- Develop a lightweight accident-warning system for real-time detection, localization, and verbal alerts; validate the system through closed-loop benchmark evaluations.

TECHNICAL SKILLS

- **Programming & ML:** Python, PyTorch, TensorFlow, JAX, Java, C#, MATLAB, R, SQL; reinforcement learning, world models, VLM/VLA models, deep learning, computer vision, and graph learning.
- **Robotics & Simulation:** ROS 2, CARLA, MetaDrive, Unity, Simulink, SUMO, VISSIM, TORCS, and Scenic.

- **Systems & Engineering:** Linux, MySQL, Flink, ADAS, hardware-in-the-loop testing, control and planning, and LiDAR/RADAR/camera sensing.

SELECTED PUBLICATIONS

First- or co-first-author publications ([†] equal contribution)

1. **Y. Qu**, Z. Sheng, Z. Huang, J. Chen et al. “Found-RL: Foundation model-enhanced reinforcement learning via asynchronous VLM feedback for autonomous driving,” *Communications in Transportation Research*.
2. **Y. Qu[†]**, Z. Huang[†], Z. Sheng[†], J. Chen et al. “VLM-SAFE: Vision-Language Model-Guided Safety-Aware Reinforcement Learning with World Models for Autonomous Driving,” *arXiv*, arXiv:2505.16377.
3. Z. Huang[†], Z. Sheng[†], **Y. Qu[†]**, J. You, and S. Chen “VLM-RL: A Unified Vision Language Models and Reinforcement Learning Framework for Safe Autonomous Driving,” *Transportation Research Part C: Emerging Technologies*.
4. **Y. Qu**, Z. Xu, Z. Huang, Z. Sheng, S. Chen, and T. Chen “MetaSSC: Enhancing 3D Semantic Scene Completion for Autonomous Driving through Meta-Learning and Long-sequence Modeling,” *Communications in Transportation Research*.
5. **Y. Qu**, Z. Li, X. Zhao, and J. Ou “Towards real-world traffic prediction and data imputation: A multi-task pretraining and fine-tuning approach,” *Information Sciences*.
6. **Y. Qu**, J. Rong, Z. Li, and K. Chen “ST-A-PGCL: Spatiotemporal adaptive periodical graph contrastive learning for traffic prediction under real scenarios,” *Knowledge-Based Systems*.
7. **Y. Qu**, Z. Li, Q. Liu, M. Pan, and Z. Zhang “Crash/Near-Crash Analysis of Naturalistic Driving Data Using Association Rule Mining,” *Journal of Advanced Transportation*.

Co-authored publications

8. Z. Sheng, Z. Huang, Y.-J. Chen, **Y. Qu**, et al. “SafePLUG: Empowering Multimodal LLMs with Pixel-Level Insight and Temporal Grounding for Traffic Accident Understanding,” *CHAIN*.
9. Z. Sheng, Z. Huang, **Y. Qu**, Y. Leng et al. “Talk2Traffic: Interactive and editable traffic scenario generation for autonomous driving with multimodal large language model,” *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*, pp. 3788–3797.
10. Z. Huang[†], Z. Sheng[†], Z. Wan[†], **Y. Qu**, et al. “Sky-Drive: A Distributed Multi-Agent Simulation Platform for Socially-Aware and Human-AI Collaborative Future Transportation,” *Journal of Intelligent and Connected Vehicles*.
11. Z. Sheng[†], Z. Huang[†], **Y. Qu**, Y. Leng, S. Bhavanam, and S. Chen “CurricuVLM: Towards Safe Autonomous Driving via Personalized Safety-Critical Curriculum Learning with Vision-Language Models,” *Transportation Research Part C: Emerging Technologies*.

SELECTED PRESENTATIONS

1. **Y. Qu** and Z. Li, “TSS-FUS: A Temporal-Spatial-Semantic Fusion Network for Area-Wide Traffic Forecasting,” *103rd Transportation Research Board Annual Meeting*.
2. **Y. Qu**, J. Chen, Z. Huang, Z. Sheng, Y. Chen, S. Chen, and S. Labi, “Towards Safe Autonomous Driving: A Dual-Phase Framework with Vision-Language Models & Human Feedback,” *International Conference on Transportation and Development (ICTD)*, 2025.
3. **Y. Qu**, Z. Xu, Z. Huang, Z. Sheng, S. Chen, and T. Chen, “MetaSSC: Enhancing 3D Semantic Scene Completion for Autonomous Driving through Meta-Learning and Long-sequence Modeling,” *104th Transportation Research Board Annual Meeting*.
4. **Y. Qu[†]**, Z. Huang[†], Z. Sheng[†], J. Chen, S. Chen, and S. Labi, “VL-SAFE: Vision-Language Guided Safety-Aware Reinforcement Learning with World Models for Autonomous Driving,” *11th Midwest Workshop on Control and Game Theory*.

SELECTED AWARDS

Lyles Teaching Assistant	Purdue University	2025–2026
First-Class Academic Scholarship	Beijing University of Technology	2022–2024
Outstanding Graduate Award	Beijing University of Technology	2023–2024
Ren Futian Transportation Scholarship	Beijing University of Technology	2023–2024
First Prize, National Mathematics Competition	National Mathematical Society	2020–2021

PROFESSIONAL SERVICE

- **Reviewer:** 10+ international journals.
- **Committee Member:** AI in Transportation Committee, American Society of Civil Engineers (ASCE).